

LESSON 7-2 • SESSION 1 • PRACTICE • FORM A

Rational Numbers on a Number Line

Learning target – I can plot a rational number and its opposite on a number line. · 6.NS.C.6c

REMEMBER

- A **rational number** can be written as a fraction of two integers: $3/4$, $-2.5 = -5/2$, $7 = 7/1$.
- To plot one, mark the integers, then split each gap into **equal parts** (halves, fourths, tenths).
- Negative numbers sit **left of 0**, or **below 0** on a vertical number line.

WORKED EXAMPLEPlot 1.5 and -3.5 on a number line.**Step 1** Mark the integers from -4 to 4 .**Step 2** Split each gap into halves.**Step 3** 1.5 is halfway between 1 and 2. -3.5 is halfway between -4 and -3 .**Answer** 1.5 is right of 0; -3.5 is left of 0**WORDS** rational number (*número racional*) · integer (*número entero*) · number line (*recta numérica*)**PRACTICE**

show your work · label units

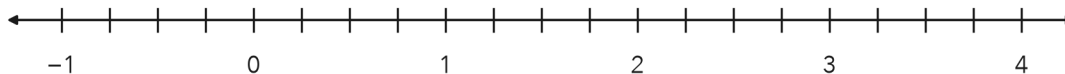
1. Name the two integers each number is between.

a. $-9/2$ _____

b. 0.75 _____

c. $-11/4$ _____d. -2.4 _____

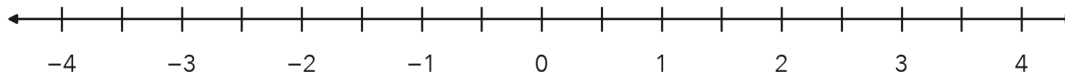
- 2.
- Garden Log.**
- A garden log measures from the soil surface (0): sunflower 3.5 m up, carrot root 0.75 m down, bean sprout 0.5 m up, tulip bulb 0.2 m down.



a. Write a rational number for each: sunflower ____ carrot ____ bean ____ tulip ____

b. Plot and label all four on the number line.

- 3.
- Lab Thermometer.**
- A lab thermometer read
- -2.5
- degrees at 6 a.m. and 2.5 degrees at noon. Plot both readings. How far is each reading from 0?

Hint Count the halves from 0 to each point.**Answer** _____ degrees

- 4.
- Mixing Bowl.**
- A cook added
- $5/4$
- cups of broth to a bowl, then removed
- $5/4$
- cups. Write the removal as a rational number in decimal form.

Hint Removing makes the number negative. $5/4 = 1\frac{1}{4}$.

Show your work.

Answer _____ cups

- 5.
- Rational Claim.**
- Lena says 7 is not a rational number because it is not written as a fraction. Write 7 as a fraction with a denominator of 1. Is Lena right?

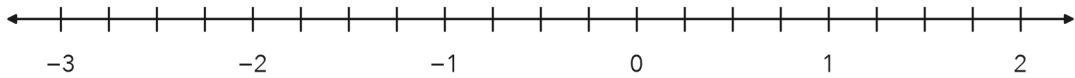
Hint Any integer can be divided by 1.

Show your work.

Fraction _____

- 6. Three Marks.** A number strip shows the marks $-1\frac{1}{4}$, 1.3 and -0.45 . Plot all three. Which mark is closest to 0?

Hint Write $-1\frac{1}{4}$ as a decimal first: -2.75 .



Closest to 0 _____

- 7. Pool Depths.** A pool's surface is 0. A float sits 0.4 m above the surface and a drain sits at -1.6 m. How many meters farther from the surface is the drain than the float?

Hint Find each distance from 0, then subtract.

Show your work.

Answer _____ m

- 8. Ledge and Cave.** A rock ledge is 12.5 meters above the sea surface. A cave floor sits at the opposite value.

- a.** Write the number for the cave floor. _____ m
- b.** Between which two integers is the cave floor? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Rafael's Work.** To plot -3.4 , Rafael counted four tenths to the RIGHT of -3 and put his point between -3 and -2 . Explain his mistake. Then name the two integers -3.4 is between.

TEST PRACTICE

choose the best answer

- 10.** On a number line, -2.5 sits between –
- A** -2 and -1 **B** -3 and -2 **C** 2 and 3 **D** -2.5 and -2
- 11.** Which number is farthest left on a number line?
- A** -4.5 **B** -1.5 **C** 0 **D** 2.5

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

2.5 N	-12.5 R	7/1 M	1.25 Q	1.2 E	12.5 X	-0.45 B	-2.5 Z	-1.25 U	0.45 J
#3		#4	#5	#6	#7	#8a			

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____